

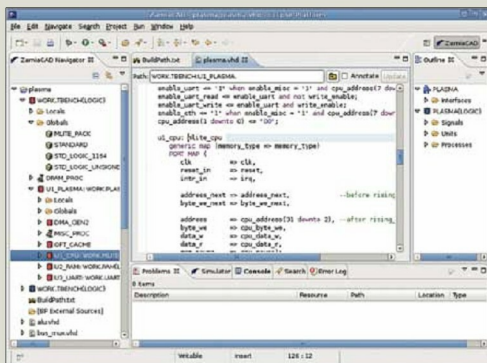


Fig. 3: Navigating through zamiaCAD

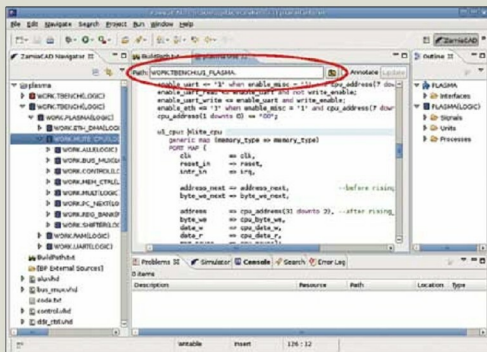


Fig. 4: Path-based editor navigation

in which files. The actual VHDL parser is then used in an on-demand manner only on local source files (those which reside in your eclipse workspace) and files that define design units that are actually needed during elaboration.

zamiaCAD's design principles

zamiaCAD is non-invasive and follows certain design principles. Consider a group of four members

working on a project. Here it is not necessary for all the members of the team to use zamiaCAD. Since the data format is VHDL, which is the language used in almost every design software, even if one member of the team is using zamiaCAD, he or she can work on the pre-existing projects, which have been created with any other design software, with minimal configuration effort.

zamiaCAD is also scalable. Persis-

Features

zamiaCAD comprises three parts: Frontend-HDL parsing; Core-intermediate design representation and analysis; and GUI-Eclipse IDE plugin.

Frontend: HDL parsing

- Full VHDL 2003 parser, syntax tree, elaboration
- Verilog 2005 parser and syntax tree, no elaboration yet
- Persistent and scalable syntax tree storage

Core: Intermediate design representation and analysis

- Based on a powerful, persistent and scalable design database
- Fully elaborated design model
- Full source back-annotation
- Static analysis (e.g., global signal reference search, fsm recognition)
- Interpreter for quick expression evaluation
- Waveform file import modules
- Built-in simulator for ad-hoc validation

GUI: Eclipse IDE plugin

- Graphical viewers and editors
- Automatic model builder

Latest Version: zamiaCAD 0.11.1 (released on January 16, 2013)

What's new?

In addition to code polishing and simulator bug fixes, an experimental scientific feature has been added to measure the quality of the diagnostic test set. Convenient zamiaCAD.bat script is now also available for running zamiaCAD from command line in Windows OS.

zamiaCAD features all-new enriched static analysis and automated VHDL design bugs localisation facility through signal assignment search. New tutorials demonstrate the debugging facilities powered by Python scripts. It also includes zamiaCAD speed-up and improvements in the simulator and editor/navigation parts.

tent data structures and caches allow processing of very large designs. It is also robust—JUnit tests exist for many parts of the Java framework.

Use areas

Four main application areas of this CAD software are design entry, design analysis, design integration and simulation. In these areas,